UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

SUBJECT: COMPUTER NETWROKS

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LAB 13

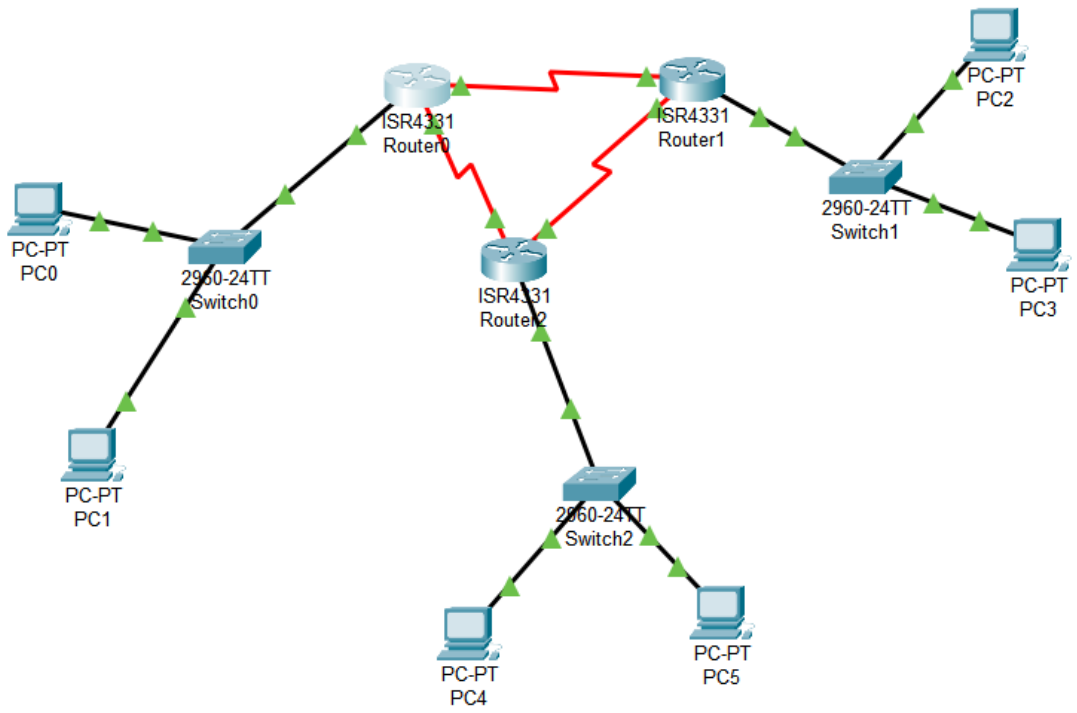
Lab 13 – Configuring OSPF in Packet Tracer

Introduction

In this lab, we learned how to configure the Open Shortest Path First (OSPF) protocol in Cisco Packet Tracer. OSPF is a link-state routing protocol used for efficient path selection and dynamic routing. By configuring OSPF on multiple routers, computers from different subnets can communicate with each other.

Network Setup

The network topology is reused from Lab-12, consisting of three routers (Router0, Router1, and Router2), each connected to its respective subnet and interconnected via serial interfaces.



OSPF Configuration

Router0 Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.3168.1.0 0.255.255.255 area 1
                             ^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.1.0 0.255.255.255 area 1
Router(config-router)#network 192.168.5.0 0.255.255.255 area 1
Router(config-router)#network 192.168.6.0 0.255.255.255 area 1
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router1 Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 2
% Unknown command or computer name, or unable to find computer address

Router(config)#
Router(config)#router ospf 2
Router(config-router)#network 192.168.2.0 0.255.255.255 area 1
Router(config-router)#
00:52:41: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.6.1 on Serial0/2/0 from LOADING to FULL, Loading
Done

Router(config-router)#network 192.168.5.0 0.255.255.255 area 1
Router(config-router)#network 192.168.7.0 0.255.255.255 area 1
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router2 Configuration

Router2

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 3
Router(config-router)#network 192.168.3.0 0.255.255.255 area 1
Router(config-router)#network 192.168.6.0 0.255.255.255 area 1
Router(config-router)#network 192.168.7.0 0.255.255.255 area 1
Router(config-router)#exit
```

Testing Connectivity

After configuring OSPF, we verified the network connectivity by pinging a computer from one subnet to another. For example, from PC0 (192.168.1.2) to PC5 (192.168.3.2). The ping was successful, confirming that OSPF is correctly routing the packets between different subnets.

PINGING PC5 From pc1

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=18ms TTL=126
Reply from 192.168.3.3: bytes=32 time=8ms TTL=126
Reply from 192.168.3.3: bytes=32 time=9ms TTL=126
Reply from 192.168.3.3: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 18ms, Average = 10ms

C:\>
```

Conclusion

This lab demonstrated how to configure OSPF on multiple routers in Packet Tracer. We observed successful communication between computers across different subnets, proving the effectiveness of OSPF in dynamic routing environments.